

Declarations

Competing Interests

The authors declare no competing interests.

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CRedit Author Contributions

Zhongyuan Liu: Conceptualization, Methodology, Software, Formal analysis, Validation, Investigation, Data curation, Writing - original draft, Writing - review and editing.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work, the author(s) used Codex, an OpenAI coding agent, to assist with data-analysis code generation, statistical sensitivity analyses, literature traceability, and manuscript editing. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data Statement

All source cohorts are public GEO series: GSE70362, GSE23130, GSE186542, GSE167199, GSE146904 and GSE207176. Derived analysis tables and the audit code are available at <https://github.com/wdsve/ivd-repro-audit> and archived at <https://doi.org/10.5281/zenodo.23133983>. The manuscript uses no private patient-level data.

Ethics Statement

This study used publicly available de-identified datasets and did not involve new recruitment or intervention.